

ROADOCC: LEARNING WHEN TO PERSIST, TRANSPORT, OR REFRESH MEMORY FOR ROADSIDE OCCUPANCY PREDICTION

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ABSTRACT

3D semantic occupancy prediction has advanced rapidly for ego-centric driving, while camera-only roadside occupancy remains underexplored. Unlike moving ego sensors, fixed roadside cameras observe a persistent world frame in which static structure remains aligned but sparse traffic moves. This asymmetry makes history valuable for weak or occluded actors but dangerous when moving evidence remains at stale locations. Motion-guided retrieval can relocate historical features, but the resulting address alone does not indicate whether the retrieved history is valid for the current voxel. We therefore formulate temporal memory as soft routing among evidence sources before fusion. For each selected voxel, RoadOcc can *persist* with fixed-coordinate history, *transport* velocity-addressed history, or *refresh* from current evidence when historical support is unreliable. Dynamic-aware Cross-attention (DCA) localizes where temporal reasoning is needed, multi-scale voxel velocity estimation (VVE) combines current and temporal motion cues to determine where history should be read, and velocity-guided dynamic sparse fusion (VDSF) routes fixed-coordinate, transported, or current evidence before fusion. Together, they separate source localization, address construction, and evidence admission. On InfraOcc, RoadOcc reaches 65.29 mIoU and 32.37 dynamic mIoU, outperforming STCOcc by 4.44 and 4.71 points. Controlled transfer and ablations further support the routing design. Code will be released.

1 INTRODUCTION

Semantic occupancy prediction reconstructs free space, static layout, and traffic participants in a common voxel volume. Camera-only methods have made this representation practical through sparse voxel queries, BEV lifting, tri-plane features, and unified panoptic volumes (Cao & de Charette, 2022; Li et al., 2023a; Wang et al., 2023; Tian et al., 2023; Li et al., 2025b; Huang et al., 2023; Wei et al., 2023; Tang et al., 2024; Huang et al., 2024b). However, these methods are predominantly developed for moving platforms, where each observation is interpreted in an ego-centric frame. A roadside sensor instead repeatedly observes a persistent scaffold overlaid by sparse, short-lived dynamics. History is most valuable for weak or occluded actors, yet stale footprints can overwrite exactly these targets. The fixed world frame therefore shifts the central temporal problem to local evidence management. Static history is already aligned, whereas dynamic history must be selectively relocated or rejected.

The same history can provide either valid support or stale evidence. Fixed-coordinate memory preserves static layout and can recover weak or occluded vehicle boundaries, but it does not relocate an object that moved within the fixed coordinate system. Reading such memory from the former footprint can blur a dynamic boundary, duplicate an object, or attach it to nearby static layout. Discarding history avoids these errors but also removes useful support. The challenge is therefore not simply whether to use history, but to decide where temporal evidence is needed, where supporting history should be read, and whether that source remains admissible. These cases can coexist within one scene or along a single object boundary, making a uniform temporal fusion rule insufficient.

Recent temporal methods demonstrate the value of multi-frame context for occupancy, flow, and 3D perception (Ma et al., 2024a; Zheng et al., 2024; Wang et al., 2025; Chen et al., 2025a; Liao

et al., 2025; Liu et al., 2025; Huang & Huang, 2022; Chang et al., 2024; Leng et al., 2025). Existing designs already incorporate temporal memory, motion alignment, and adaptive fusion. However, they largely treat history as a feature to align and weight. Motion alignment resolves displacement but not evidence provenance. A flow-proposed address may still lack class-consistent support for the current voxel. A stationary object may reuse fixed-coordinate memory, a moving object may retrieve memory near $\mathbf{x} - \Delta t \mathbf{v}_t(\mathbf{x})$, and an appearing, occluded, or historically unsupported object must instead rely on current evidence. Continuous feature weighting can represent these alternatives, but existing formulations typically leave the rejection of unsupported transported history implicit.

We therefore formulate temporal occupancy as explicit evidence-source routing. *Should each selected voxel Persist, Transport, or Refresh?* Persist reuses the fixed-coordinate source, Transport applies backwarp sampling to velocity-addressed history, and Refresh uses no history, only current evidence. Flow proposes where history should be read, while P/T/R determines which of these sources is semantically admissible. This separation makes motion correction and source validation independently testable.

RoadOcc realizes this principle through a DCA–VVE–VDSF chain. DCA localizes the dynamic candidate and refines the current observation. VVE then combines current and temporal flow into dynamic flow, after which VDSF selects fixed-coordinate, backwarped, or current evidence. The stages therefore resolve where temporal reasoning is needed, where history should be read, and which source should enter fusion. Their intermediate outputs also expose whether an error originates from dynamic localization, historical addressing, or source selection instead of hiding all three effects inside an opaque fusion block. Figure 1 grounds the three evidence-source alternatives. Our contributions are summarized below.

- We propose RoadOcc, which formulates roadside temporal occupancy as a Persist–Transport–Refresh routing problem. By separating motion address from source admissibility, it retains useful history while preventing stale or unsupported evidence from entering fusion.
- We realize this principle with a staged DCA–VVE–VDSF chain that sequentially resolves where history is needed, where it should be read, and whether it is admissible. DCA localizes dynamic candidates, VVE estimates multi-scale dynamic flow, and VDSF performs P/T/R source selection before fusion.
- We evaluate the source-routing hypothesis with dynamic-first occupancy, flow-advected next-frame consistency, and velocity diagnostics. Controlled ablations separate address correction from admissible source selection and test whether both decisions explain occupancy gain.

2 RELATED WORK

2.1 CAMERA OCCUPANCY AND ROADSIDE PERCEPTION

Camera occupancy methods lift multi-view image evidence into volumetric representations. MonoScene Cao & de Charette (2022) and VoxFormer Li et al. (2023a) establish image-based semantic scene completion, while BEVDet, BEVDepth, and BEVFormer improve multi-

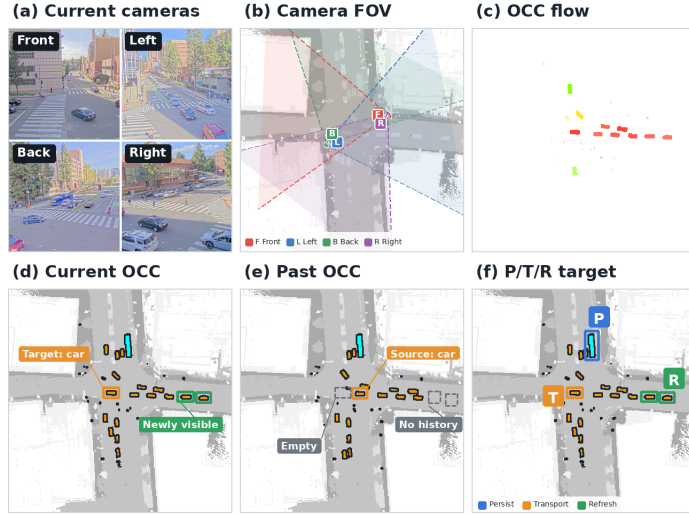


Figure 1: **Real-scene evidence for explicit source routing.** The top row shows current views, camera FOVs, and current-aligned occupancy flow (a–c). The bottom row shows current/past occupancy and P/T/R targets (d–f), illustrating a persistent bus, transported car, and newly visible cars assigned Refresh.

view lifting and BEV reasoning (Huang et al., 2021; Li et al., 2023b; 2025b). Occupancy-specific designs span tri-plane, dense, sparse, Gaussian, and forward-backward representations (Huang et al., 2023; Wei et al., 2023; Zhang et al., 2023; Tang et al., 2024; Huang et al., 2024b; Li et al., 2023c), as well as panoptic and instance-aware prediction (Wang et al., 2024; Li et al., 2025a). Unlike MC-BEVRO’s roadside BEV occupancy monitoring (Vaghela et al., 2025), RoadOcc routes evidence for class-aware 3D occupancy.

InfraOcc provides a roadside occupancy benchmark emphasizing persistent infrastructure (Yang et al., 2026). Figure 2 reveals its temporal asymmetry. Fixed roadway structure provides broad, stable support, while dynamic participants form sparse, short-lived trails. This asymmetry favors retaining static context while selectively relocating moving evidence.

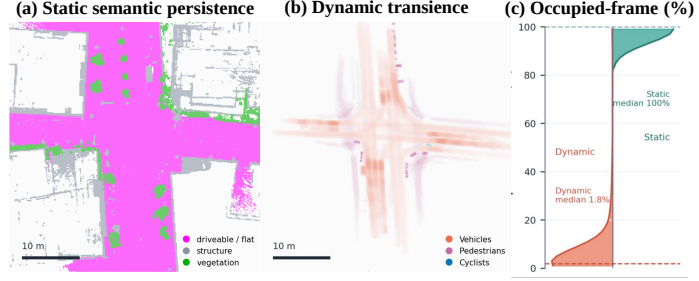


Figure 2: **Static-to-dynamic persistence in InfraOcc.**

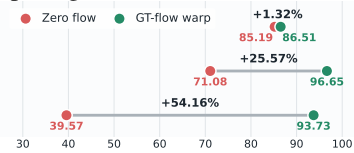
2.2 TEMPORAL OCCUPANCY, MOTION, AND ALIGNMENT

Temporal occupancy methods improve memory through recurrent aggregation, motion-aware alignment, and explicit association. Recurrent and video-based models maintain or forecast scene state (Huang & Huang, 2022; Chang et al., 2024; Huang et al., 2024a; Mohan et al., 2026), while occupancy-flow methods couple volumetric semantics with motion (Ma et al., 2024a; Zheng et al., 2024; Wang et al., 2025; Chen et al., 2025a; Liao et al., 2025; Liu et al., 2025; Murhij & Yudin, 2024; Zheng et al., 2026). Other approaches organize temporal cues, learn semantic associations, or retrieve traversal priors (Leng et al., 2025; Chen et al., 2025b; Ouyang et al., 2025; Yuan et al., 2025). Across these designs, history is aligned, associated, forecast, or adaptively weighted, but whether the retrieved source remains semantically valid is left implicit. RoadOcc makes this subsequent source-admissibility choice explicit among fixed-coordinate history, transported history, and current evidence.

Table 1: **Generated GT-flow target at $t + 0.5$ s.** Zero-flow and GT-flow transport share the inter-frame transform and differ only by object displacement. Same/Dyn./Occ. denote unchanged semantics, dynamic class, and occupied class. Values are percentages and N counts valid queries in millions.

Speed	N (M)	Zero-flow transport			GT-flow warp			Δ Same
		Same	Dyn.	Occ.	Same	Dyn.	Occ.	
All	11.83	72.93	78.86	90.92	89.21	95.79	99.38	+16.28
Slow (< 2 m/s)	7.87	85.19	94.08	98.80	86.51	96.25	99.70	+1.32
Med. (2–5 m/s)	1.16	71.08	71.38	87.50	96.65	97.64	99.53	+25.57
Fast (≥ 5 m/s)	2.80	39.57	39.59	70.41	93.73	93.75	98.43	+54.16

Figure 3: **Motion compensation restores temporal consistency.** GT flow preserves dynamic support across speed ranges, whereas fixed-coordinate retrieval becomes increasingly misaligned as speed grows.



Flow-based alignment is closest to this problem. AsyncBEV aligns asynchronous cross-modal observations with BEV feature flow (Wang et al., 2026). In the fixed roadside setting, Table 1 and Figure 3 show that GT-flow warping improves same-class agreement by 1.32 points for slow objects, but by 25.57 and 54.16 points for medium and fast objects. The rapidly growing gap shows that fixed-coordinate reuse fails where displacement makes recovery most important, motivating selective motion compensation.

3 METHOD

3.1 PRELIMINARY ANALYSIS

Let $\mathcal{I}_t = \{I_t^n\}_{n=1}^4$ denote the synchronized roadside images at time t . RoadOcc predicts semantic occupancy $\mathbf{y}_t$ over a voxel domain Ω and a planar velocity field $\mathbf{v}_t : \Omega \rightarrow \mathbb{R}^2$. Among the 18 occu-

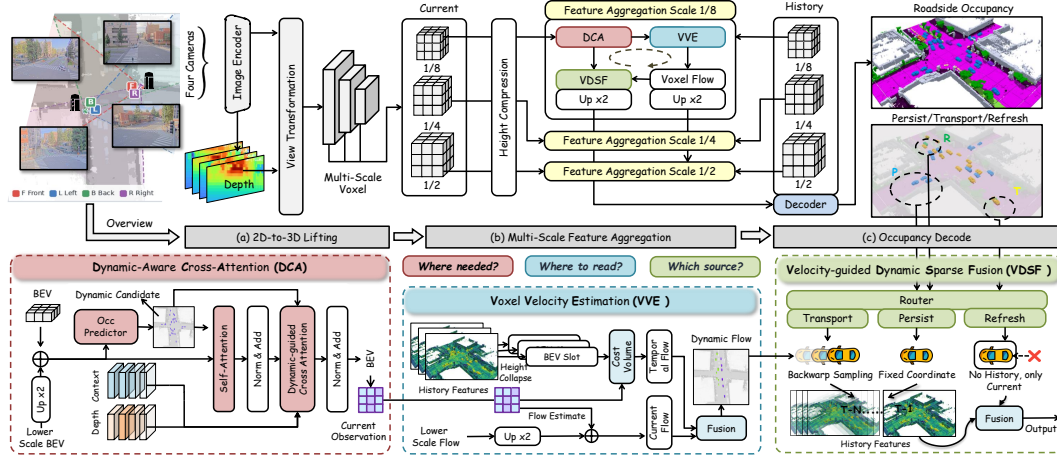


Figure 4: **RoadOcc framework.** The image encoder and view transformation construct multi-scale voxel features. At each feature aggregation scale, DCA produces the current observation, VVE fuses current, temporal, and lower-scale flow into dynamic flow, and VDSF routes fixed-coordinate, backwarped, or current evidence to fusion. Coarse-to-fine outputs feed occupancy decoding for semantic occupancy and voxel flow.

pancy states, the dynamic set is $\mathcal{C}_{\text{dyn}} = \{\text{bicycle, bus, car, motorcycle, pedestrian, truck}\}$. Velocity is expressed in the common roadside horizontal frame.

Because fixed roadside cameras share a persistent physical frame, cached history already lies on the current voxel grid at inference. We only reconcile relative BEV augmentations during training. Under a locally constant velocity approximation, history supporting a current voxel $\mathbf{x}$ is read from

$$\tilde{\mathbf{h}}_{t-\Delta t}(\mathbf{x}) = \text{Sample}(\bar{\mathbf{h}}_{t-\Delta t}, \mathbf{x} - \Delta t \mathbf{v}_t(\mathbf{x})), \quad (1)$$

where $\bar{\mathbf{h}}$ is same-grid history and Sample is trilinear interpolation. The three candidates are fixed-coordinate history at $\mathbf{x}$ (Persist), flow-addressed history at $\mathbf{x} - \Delta t \mathbf{v}_t(\mathbf{x})$ (Transport), and current evidence (Refresh). Equation 1 specifies only an address. VDSF still decides whether the sampled source has class-consistent historical support.

3.2 OVERVIEW

RoadOcc decodes $\{\mathbf{V}_t^s\}_{s \in \{1/8, 1/4, 1/2\}}$ from coarse to fine. As shown in Figure 4, the encoder lifts four camera views into multi-scale voxel features. At each aggregation stage, DCA determines where evidence is needed, VVE determines where history should be read, and VDSF determines which source should enter fusion. The output feature and dynamic flow seed the next finer stage before occupancy decoding. Supplementary Figure A4 visualizes this progression.

3.3 DYNAMIC-AWARE CROSS-ATTENTION (DCA)

Dynamic objects occupy few voxels, so uniformly repeating image cross-attention spends most computation on stable structure. After the first nonempty-guided image update, DCA combines the current representation with an upsampled lower-scale BEV to derive a dynamic candidate. Let $\pi_{t,\text{cur}}^s$ and $\pi_{t,\text{sta}}^s$ denote the provisional distributions produced by the current and static prediction heads after the first image update, and let $\mathcal{C}_{\text{sta}}^+$ contain the static classes and free space. The mean semantic discrepancy $\delta_{t,\text{sta}}^s$ over $\mathcal{C}_{\text{sta}}^+$ and current dynamic probability $p_{t,\text{dyn}}^s$ form the candidate map

$$c_t^s(\mathbf{x}) = \text{Norm}(\max\{\delta_{t,\text{sta}}^s(\mathbf{x}), p_{t,\text{dyn}}^s(\mathbf{x})\}). \quad (2)$$

Here Norm divides by the per-sample spatial maximum, clamped to 10^{-6} . Discrepancy recovers departures from the static hypothesis, while dynamic probability retains visible actors. Their maximum also preserves early semantic misses detected through change evidence.

The candidate map controls a second image cross-attention rather than directly reweighting the voxel feature. Following projected deformable attention (Li et al., 2025b) and occupancy-aware reference

sampling (Liao et al., 2025), let $\mathbf{q}_{t,\mathbf{x}}^s$ be the BEV query at $\mathbf{x}$, $\mathcal{R}_n(\mathbf{x})$ its visible height anchors in camera n , and $\mathbf{F}_t^n$ the corresponding image feature. DCA retains $\mathcal{R}_{n,c}^s(\mathbf{x}) = \{\mathbf{r} \in \mathcal{R}_n(\mathbf{x}) : u_{\mathbf{r}} < c_t^s(\mathbf{r})\}$ and updates the query as

$$\mathbf{q}_{t,\mathbf{x}}^{s,+} = \mathbf{q}_{t,\mathbf{x}}^s + \mathbf{W}_o \frac{1}{|\mathcal{V}_{\mathbf{x}}|} \sum_{n \in \mathcal{V}_{\mathbf{x}}} \sum_{\mathbf{r} \in \mathcal{R}_{n,c}^s(\mathbf{x})} \beta_{n,\mathbf{r}}^s \mathcal{A}_{\text{def}}(\mathbf{q}_{t,\mathbf{x}}^s, \mathcal{P}_n(\mathbf{r}), \mathbf{F}_t^n), \quad (3)$$

where $\mathbf{r} = (\mathbf{x}, z_{\mathbf{r}})$ indexes a voxel-height score shared across cameras, $\mathcal{V}_{\mathbf{x}}$ contains valid cameras, and $\mathcal{P}_n$ projects anchors. Depth consistency $\beta_{n,\mathbf{r}}^s$ multiplies normalized deformable weights before camera averaging. During training, $u_{\mathbf{r}}$ is sampled from a normal distribution with mean 0.5 and standard deviation 1, truncated to $[0, 1]$. At inference, it is fixed to 0.5. The stochastic rule exposes uncertain boundary voxels during training, while deterministic thresholding keeps inference stable. Queries without retained support follow the residual path unchanged.

The candidate gates a second cross-attention over image depth context and also allocates VDSF tokens. DCA therefore localizes where additional image and temporal evidence is needed without repeatedly updating the stable scene.

3.4 MULTI-SCALE VOXEL VELOCITY ESTIMATION (VVE)

DCA refines the current observation but does not locate displaced history. VVE estimates its address from the DCA feature and nearest valid history. From the current occupancy distribution $\pi_t^s(\mathbf{x})$, it computes dynamic support $d_t^s = \sum_{c \in \mathcal{C}_{\text{dyn}}} \pi_{t,c}^s$ and nonempty support $e_t^s = 1 - \pi_{t,\text{free}}^s$. Here d_t^s guides height compression into the current BEV slot $\mathbf{B}_t^s$, while e_t^s guides temporal token selection. Applying the same compression to history gives $\bar{\mathbf{B}}_{t-\Delta t}^s$.

VVE next compares each current BEV slot with a small neighborhood in the historical slot, producing the local cost volume

$$\mathbf{K}_t^s(\mathbf{x}, \delta) = \left\langle \frac{\mathbf{B}_t^s(\mathbf{x})}{\|\mathbf{B}_t^s(\mathbf{x})\|_2}, \frac{\bar{\mathbf{B}}_{t-\Delta t}^s(\mathbf{x} + \delta)}{\|\bar{\mathbf{B}}_{t-\Delta t}^s(\mathbf{x} + \delta)\|_2} \right\rangle, \quad \delta \in [-r, r]^2, \quad (4)$$

where $r = 2$ yields a 5×5 neighborhood and both norms use an $\epsilon = 10^{-6}$ floor. The cost-volume decoder converts this local correspondence evidence into temporal flow.

VVE estimates flow recursively rather than solving each resolution in isolation. Let s^- denote the preceding coarser stage and set the upsampled lower-scale flow $\mathbf{v}_{t,\uparrow}^s$ to zero at $1/8$ and to $\text{Up}_s(\mathbf{v}_{t,\text{cur}}^{s-})$ otherwise. This prior initializes the current flow. Current appearance and its semantic embedding $\mathbf{S}_t^s$ predict a current-only correction, while the history feature and cost volume predict the temporal flow. Their fusion is

$$\begin{aligned} \mathbf{v}_{t,\text{cur}}^s &= \mathbf{v}_{t,\uparrow}^s + \Delta \mathbf{v}_{t,\text{cur}}^s, & \mathbf{v}_{t,\text{tmp}}^s &= f_{\text{tmp}}(\mathbf{B}_t^s, \bar{\mathbf{B}}_{t-\Delta t}^s, \mathbf{K}_t^s), \\ \hat{\mathbf{v}}_t^s &= (1 - \alpha_t^s) \mathbf{v}_{t,\text{cur}}^s + \alpha_t^s \mathbf{v}_{t,\text{tmp}}^s + \Delta \mathbf{v}_{t,\text{ref}}^s. \end{aligned} \quad (5)$$

Here f_{tmp} is the cost-volume decoder and $\alpha_t^s = 0.5 \mathbf{1}_{\text{hist}} \max_z d_t^s$ gates history, becoming zero at sequence starts. The residual $\Delta \mathbf{v}_{t,\text{ref}}^s$ produces the dynamic flow, which is lifted across height as the VDSF address. Coarse stages capture motion structure and fine stages refine boundaries.

3.5 VELOCITY-GUIDED DYNAMIC SPARSE FUSION (VDSF)

Velocity-guided sparse fusion. Applying the full temporal memory after DCA and VVE would mostly repeat static context. VDSF makes gather–fuse–scatter (Liao et al., 2025) motion aware by combining DCA guidance with nonempty support under a fixed token budget

$$\mathbf{g}_t^s = \text{clip}(c_t^s + \eta e_t^s, 0, 1), \quad \mathcal{X}_t^s = \text{TopK}(\mathbf{g}_t^s), \quad (6)$$

where η balances change evidence and occupied support. The selected set $\mathcal{X}_t^s$ receives full stage-local history. Its complement uses a short fixed-coordinate path, preserving background context at lower cost.

Each scale receives multi-frame history features and their VVE motion priors. For $\mathbf{x} \in \mathcal{X}_t^s$, elapsed time converts dynamic flow into a voxel offset. Backwarp sampling produces the Transport candidate $\mathbf{h}^T$, while the fixed-coordinate read produces the Persist candidate $\mathbf{h}^P$.

Table 2: Main InfraOcc occupancy results.

Method	gIoU	mIoU			Class																	
		all	dyn	sta	Bike	Bus	Car	Motorcycle	Pedestrian	Truck	Other	Barrier	Cone	Driveable	Sidewalk	Terrain	Manmade	Vegetation	Free			
																						
BEVDet '21	68.62	45.14	21.19	63.11	9.23	42.75	38.04	9.68	15.01	12.45	42.88	70.36	53.77	72.04	68.07	73.09	64.28	60.38	94.87			
BEVFormer '22	81.72	47.76	10.73	75.53	7.97	20.39	17.65	3.47	8.62	6.30	42.47	76.26	72.83	85.19	81.95	88.95	79.52	77.05	97.56			
BEVDepth '23	71.89	46.81	19.64	67.19	14.84	33.87	38.74	3.42	16.86	10.14	48.73	71.05	62.37	73.31	69.32	76.07	70.51	66.15	95.54			
TPVFormer '23	73.80	43.09	4.36	72.13	0.10	3.69	13.80	0.41	6.86	1.33	42.86	75.29	69.69	80.61	77.62	86.63	72.61	71.70	96.12			
SurroundOcc '23	83.78	50.77	11.37	80.32	12.66	21.90	18.54	2.06	9.50	3.55	65.09	79.04	72.70	86.38	83.99	91.10	83.16	81.09	97.80			
CONet '23	78.56	49.25	18.93	72.00	3.06	37.24	39.47	4.42	17.64	11.72	45.01	75.31	64.94	84.29	79.45	84.27	72.96	69.79	97.08			
ALOCC '25	75.66	43.31	5.86	71.40	8.41	4.04	8.52	0.00	4.48	9.71	63.90	68.97	63.40	90.09	77.77	78.16	65.07	63.85	97.12			
Let Occ Flow '24	87.18	51.38	8.78	83.34	11.27	11.53	14.07	4.34	8.12	3.34	62.06	75.06	85.56	92.55	85.63	92.98	87.10	85.74	98.56			
SparseOcc '24	71.61	45.42	9.41	72.42	1.50	20.94	24.87	0.00	4.60	4.57	67.68	76.37	61.51	74.67	76.59	88.54	68.79	65.19	96.31			
GaussianFormer '24	77.89	43.63	10.45	68.51	3.13	22.11	23.17	1.29	8.21	4.80	36.70	68.54	55.60	82.59	79.37	74.20	77.85	73.21	96.71			
CRT-Fusion-S '24	88.73	55.94	21.98	81.40	13.02	45.16	38.48	7.34	18.99	8.90	63.53	71.99	74.82	93.38	84.40	91.19	86.39	85.53	98.73			
STCOcc-S '25	78.54	50.71	23.48	71.14	13.43	47.83	39.62	10.21	17.83	11.93	50.18	71.54	58.33	90.30	78.78	84.90	67.72	67.32	97.47			
CRT-Fusion '24	80.35	53.83	27.02	73.94	32.58	46.63	39.49	12.22	20.81	10.41	53.08	72.27	68.15	89.06	79.09	84.48	72.06	73.30	97.68			
STCOcc '25	92.54	60.85	27.66	85.74	27.13	48.78	44.93	10.23	22.85	12.03	68.84	70.93	81.60	95.88	89.41	95.36	91.85	92.04	99.18			
RoadOcc (ours)	94.03	65.29	32.37	89.98	37.60	55.46	49.36	13.01	22.66	16.13	87.66	72.05	86.03	97.85	92.81	96.90	92.91	93.66	99.35			

Persist-Transport-Refresh routing. Dynamic flow proposes where to read history, but a corrected address may still lack semantic support. VDSF therefore decides which source should enter fusion. It retains $\mathbf{h}^P$ and $\mathbf{h}^T$ and defines the current-only Refresh candidate as $\mathbf{h}^R = \mathbf{q}_t$. During training, the route target is constructed from the current and nearest historical semantic occupancy labels together with the motion target, and assigns each current dynamic voxel according to class-consistent historical support:

$$z_t(\mathbf{x}) = \begin{cases} P, & \|\mathbf{v}_t(\mathbf{x})\| \leq \epsilon \wedge \text{Supp}(\mathbf{x}), \\ T, & \|\mathbf{v}_t(\mathbf{x})\| > \epsilon \wedge \text{Supp}(\mathbf{x} - \Delta t \mathbf{v}_t(\mathbf{x})), \\ R, & \text{otherwise}, \end{cases} \quad (7)$$

where Supp tests same-class occupancy at the same height within the local correspondence radius. We use $\epsilon = 10^{-3}$ m/s and a one-voxel radius to absorb discretization error. The construction follows the semantic occupancy label space used by InfraOcc and is evaluated at every execution scale using its physical cell size, with a native-grid endpoint for full-resolution supervision. Under this rule, supported stationary voxels Persist, supported moving voxels Transport, and voxels without a supported historical candidate Refresh. Future occupancy is used only for offline target construction and never enters the model. The supplement details source alignment and label generation.

At each feature aggregation scale, the router combines the current observation, nearest fixed-coordinate history feature, normalized dynamic flow, and upsampled coarser P/T/R distribution. It predicts stage-local logits. A lightweight native head upsamples the finest context with full-resolution flow. Softmax gives $\alpha_t^s = (\alpha_P, \alpha_T, \alpha_R)$ and the routed candidate

$$\mathbf{h}_t^{\text{PTR}} = \alpha_P \mathbf{h}^P + \alpha_T \mathbf{h}^T + \alpha_R \mathbf{h}^R, \quad (\alpha_P, \alpha_T, \alpha_R) = \text{softmax}(\psi(\cdot)). \quad (8)$$

P/T/R therefore compete as one route distribution rather than independent weights. Transport uses backwarp sampling, Persist uses the fixed-coordinate source, and Refresh uses no history, only current evidence. Fusion uses soft probabilities and reserves argmax for diagnostics.

Let $\mathbf{H}_t^{\text{PTR},s}(\mathbf{x})$ concatenate the routed multi-frame history at scale s . VDSF combines this temporal evidence with the current observation and its semantic embedding through the fusion MLP

$$\mathbf{q}_{t,\text{fuse}}^s(\mathbf{x}) = \mathbf{q}_t^s(\mathbf{x}) + \Phi_s \left([\mathbf{q}_t^s(\mathbf{x}), \mathbf{H}_t^{\text{PTR},s}(\mathbf{x}), \text{Emb}(\pi_t^s(\mathbf{x}))] \right). \quad (9)$$

The output residual is scattered to $\mathcal{X}_t^s$ and joined with the background path. The fused volume and dynamic flow feed the next scale and are retained for subsequent frames. Detailed memory, token, source-matching, and routing settings appear in supplementary Secs. 3–5.

3.6 TRAINING OBJECTIVE

Training jointly optimizes four functional objectives.

$$\mathcal{L} = \lambda_{\text{depth}} \mathcal{L}_{\text{depth}} + \mathcal{L}_{\text{sem}} + \lambda_{\text{mot}} \mathcal{L}_{\text{mot}} + \lambda_{\text{route}} \mathcal{L}_{\text{route}}. \quad (10)$$

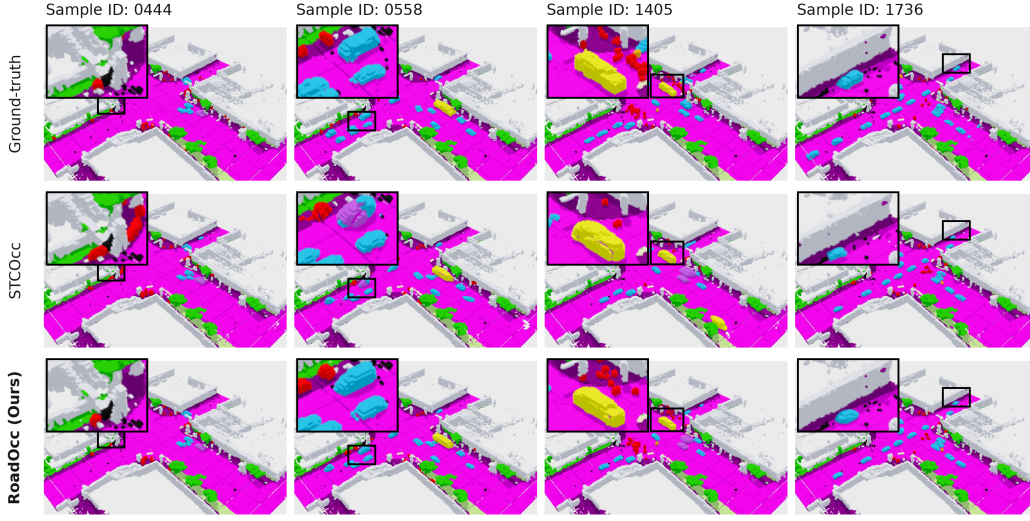


Figure 5: **Qualitative comparison with STCOcc.** RoadOcc more closely recovers dynamic-object presence, semantic identity, and geometry, including spurious pedestrian occupancy, vehicle-class confusion, bus-shape distortion, and an incomplete car footprint.

Table 3: **Joint occupancy–velocity comparison.**

Method	Occupancy					Velocity	Diagnostics	
	mIoU	Dyn.	Sta.	gIoU	RayIoU	Direct MAVE	TP-MAVE	DSR
CRT-Fusion (NeurIPS 2024)	53.83	27.02	73.94	80.35	48.08	2.227	1.839	57.56
+ VDSF P/T/R	57.15	28.30	78.79	83.46	52.34	1.990	1.958	59.45
Δ	+3.32	+1.28	+4.85	+3.11	+4.26	−0.237	+0.119	+1.89
STCOcc (CVPR 2025)	60.85	27.66	85.74	92.54	55.94	1.946	0.703	57.12
+ VDSF P/T/R	61.81	28.80	86.56	93.59	58.07	1.770	0.648	58.78
Δ	+0.96	+1.14	+0.82	+1.05	+2.13	−0.176	−0.055	+1.66
RoadOcc (ours)	65.29	32.37	89.98	94.03	65.44	1.669	<u>0.661</u>	61.11

$\mathcal{L}_{\text{depth}}$ supervises camera depth for view transformation, while $\mathcal{L}_{\text{sem}}$ covers full- and multi-scale occupancy together with the static, dynamic, and DCA auxiliaries. $\mathcal{L}_{\text{mot}}$ supervises the dynamic flow used for backwarp sampling, whereas $\mathcal{L}_{\text{route}}$ supervises the P/T/R router on valid dynamic voxels. Together, they separate where history features are read from which source enters fusion. Full target construction, weights, and schedules are given in supplementary Sec. E.

4 EXPERIMENTS

Dataset and metrics. We evaluate camera-only roadside occupancy on InfraOcc, derived from V2X-Real (Xiang et al., 2024; Yang et al., 2026). Four synchronized views are mapped to 18 occupancy states. Under the InfraOcc protocol, semantic mIoU excludes construction vehicle, trailer, other flat, and Free. Free IoU is reported separately in the class-wise table. We report semantic and geometric IoU, next-frame dynamic IoU, and motion coverage. Direct MAVE covers all GT dynamic voxels, while TP-MAVE conditions on semantic matches and is paired with recall (DSR). InfraOcc provides the dense fixed-roadside camera-only protocol used throughout. Since existing roadside datasets mainly provide object-level labels, we complement the benchmark results with matched backbone transfer and controlled studies. Full benchmark details appear in the supplement.

Implementation details. RoadOcc uses a ResNet-50 backbone, 256×704 images, and three voxel stages from 0.8 m to 0.4 m over $[-64, 64] \times [-64, 64] \times [-4.8, 1.6]$ m. Frames are 0.5 s apart. AdamW training runs for 24 epochs from 5×10^{-4} with 10^{-2} weight decay and cosine annealing. Comparisons share the voxel range, ignored classes, and evaluator. Controlled variants keep the backbone, data split, optimizer, schedule and other configuration fixed.

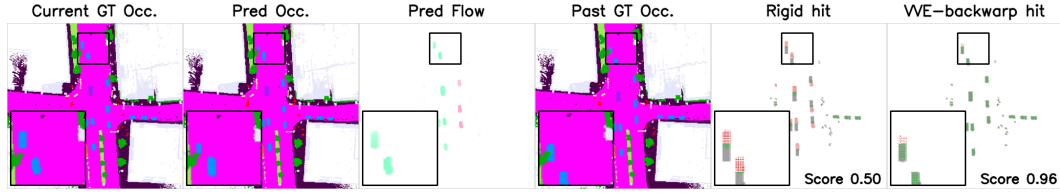


Figure 6: **Velocity-guided historical retrieval.** With queried dynamic targets fixed by current GT, the panels show the end-to-end occupancy prediction and compare a fixed-coordinate read with the VVE backtrace into aligned past GT. Black boxes are enlarged in the insets. ■ same-class matches, ■ unmatched targets, and ■ other queried voxels are shown in green, red, and gray, respectively. The score is the same-class hit rate within the one-voxel matching radius.

4.1 MAIN RESULTS

InfraOcc occupancy benchmark. Table 2 tests whether the proposed routing chain improves occupancy under the InfraOcc evaluator. RoadOcc establishes a new state of the art at 65.29 mIoU, a +4.44 gain over STCOcc. Dynamic mIoU rises by +4.71 to 32.37, while static mIoU rises by +4.24 to 89.98. The joint gain follows the framework’s staged decisions. DCA supplies the dynamic candidate, VVE converts current and temporal flow into source addresses, and the VDSF router selects among fixed-coordinate, backwarped, and current evidence. Together, they improve transient-object recovery without destabilizing the persistent roadside scaffold.

Joint occupancy-velocity behavior. Table 3 tests whether these occupancy gains follow the proposed routing mechanism. Direct MAVE measures the quality of the VVE address, while DSR measures the dynamic support recovered by the complete routing path. Relative to STCOcc, RoadOcc lowers Direct MAVE from 1.946 to 1.669 and raises DSR from 57.12 to 61.11. This paired improvement supports both routing decisions. VVE improves the dynamic flow used for backwarp sampling, while P/T/R determines which source is admissible for fusion. Adding VDSF P/T/R to CRT-Fusion and STCOcc likewise raises Dyn. by +1.28 and +1.14, respectively, while improving Direct MAVE and DSR. The consistent trend across both backbones links the gain to source-aware routing rather than backbone-specific capacity. TP-MAVE conditions on semantic matches and must be read with DSR. For CRT-Fusion, the +1.89 DSR gain indicates broader dynamic recovery even though the conditional error over matched voxels increases.

Qualitative occupancy recovery. Figure 5 supports the aggregate dynamic gain with representative matched scenes. STCOcc produces stale or spurious pedestrian occupancy, confuses a vehicle class, distorts a bus footprint, and incompletely recovers a car. By correcting displaced reads and rejecting unsupported history, RoadOcc suppresses these errors and recovers object identity and extent more consistently with GT, while preserving the surrounding roadway.

4.2 ABLATION STUDIES

Using Liao et al. (2025) as the baseline, we hold the backbone, token budget, temporal-memory configuration, and training protocol fixed unless explicitly varied. Tables 4, 5, and 6 then isolate component composition, historical-address correction, and P/T/R source selection, respectively.

Table 4: **Component ablation.**

Base	Design			Output ↑		
	DCA	VVE	VDSFP/T/R	All	Dyn.	Nxt.
✓				60.85	27.66	26.74
✓			✓	61.05	28.01	26.99
✓	✓			61.69	28.55	27.64
✓	✓		✓	63.01	30.10	29.44
✓	✓	✓	✓	64.17	30.97	30.78
✓	✓	✓	✓	65.29	32.37	31.72

Overall ablation. Table 4 follows the routing chain. VDSF alone adds only 0.20 mIoU because moving evidence remains at unresolved source locations. DCA and VDSF then raise Dyn. by +1.55 and Nxt. by +1.80, showing that the dynamic candidate identifies where sparse temporal fusion is useful. With these locations fixed, VVE adds +1.16 mIoU by correcting the dynamic flow and backwarp-sampling address. P/T/R adds a further +1.12 mIoU by selecting fixed-coordinate, backwarped, or current evidence for fusion. The ordered gains therefore track the three framework decisions rather than undifferentiated temporal capacity.

Address correction and source selection. With capacity and token budget fixed, the fixed-coordinate and VVE reads in Table 5 differ only in historical address. VVE raises Dyn. by +0.87, lowers dMAVE from 2.146 to 1.690, and raises DSR by +1.51, directly verifying address correction. With that address fixed, P/T/R adds +1.40 Dyn. and +1.76 DSR. This second transition shows that a corrected address is useful only when the fusion stage can select supported evidence.

P/T/R routing controls. Table 6 holds DCA–VVE–VDSF fixed and isolates source selection. Gate tests adaptive weighting, State introduces explicit P/T/R states, and Full uses the predicted routes in fusion. State exceeds Gate by +0.63 Dyn., +0.44 Nxt., and +0.77 DSR, while Full adds another +0.49, +0.34, and +0.59. Overall, Full surpasses Gate by +1.12 Dyn., +0.78 Nxt., and +1.36 DSR, attributing the gain to explicit source states and route-conditioned fusion rather than adaptive weighting alone.

P/T/R source-admissibility routing. P/T/R represents semantic source validity rather than instance correspondence. Persist and Transport require same-class, height-consistent support at the fixed-coordinate and VVE-addressed locations, while Refresh covers neither source. GT-conditioned diagnostics isolate routing from detection, whereas Dyn., Nxt., Direct MAVE, and DSR retain end-to-end evaluation. This construction turns source validity into a direct routing decision rather than an implicit feature weight. The supplement details construction.

Visual analysis. Figure 6 compares zero-motion reading with VVE backtracing. Current GT fixes the queried dynamic voxels and their classes, while predicted occupancy provides the end-to-end context. Red-to-green changes show dynamic flow relocating queries through backwarp sampling onto same-class past GT, isolating address quality from the subsequent routing decision.

On the same GT dynamic support, Figure 7 then shows Persist following fixed-coordinate support, Transport following displacement, and Refresh rejecting unsupported history. The paired controls therefore separate where history is read from which source is selected.

Together, the two visualizations summarize RoadOcc’s evidence-routing principle. VVE resolves *where* each queried voxel should retrieve historical evidence, while P/T/R verifies its semantic validity, preserving supported history and refreshing unreliable evidence before VDSF fusion.

5 CONCLUSION

In this work, we present RoadOcc, a voxel-level evidence-source routing framework for temporal roadside occupancy. Rather than relying on implicit temporal aggregation, RoadOcc explicitly determines where supported history should be retrieved and whether it should be reused. DCA uses current image evidence to identify motion-sensitive voxels, VVE estimates their historical source addresses, and VDSF turns these addresses into Persist, Transport, or Refresh decisions. On the InfraOcc protocol, the unified formulation achieves state-of-the-art occupancy accuracy while improving dynamic recovery and velocity estimation. Future work will extend this principle to moving platforms, longer temporal horizons, and instance-level modeling.

Table 5: **Controlled address and source selection.** All variants use the same sparse-token budget and model capacity.

Variant	Design		Output			
	VVE	P/T/R	Dyn.↑	Nxt.↑	dM↓	DSR↑
Fixed read			30.10	29.44	2.146	57.84
VVE read	✓		30.97	30.78	1.690	59.35
Full	✓	✓	32.37	31.72	1.669	61.11

Table 6: **P/T/R routing controls.** Mean $\pm$ standard deviation over three independent seeds.

Variant	Dyn.↑	Nxt.↑	DSR↑	dM↓
None	30.97 $\pm$.27	30.78 $\pm$.34	59.35 $\pm$.39	1.690 $\pm$.019
Gate	31.25 $\pm$.20	30.94 $\pm$.30	59.75 $\pm$.45	1.685 $\pm$.018
State	31.88 $\pm$.18	31.38 $\pm$.25	60.52 $\pm$.40	1.675 $\pm$.017
Full	32.37$\pm$.11	31.72$\pm$.26	61.11$\pm$.38	1.669$\pm$.020

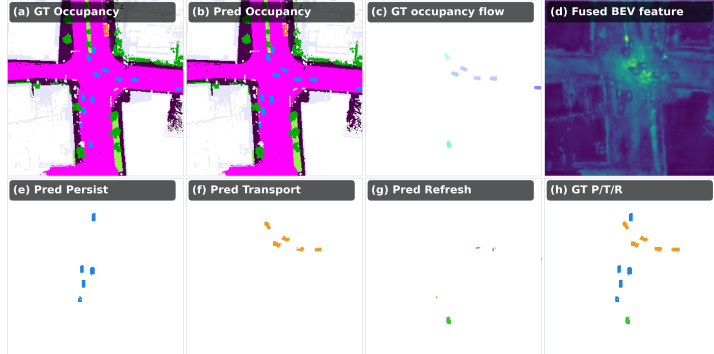


Figure 7: **Predicted P/T/R on GT dynamic support.**

AI USE STATEMENT

Generative AI tools were used for literature organization, reviewer-style feedback, and language editing. The authors verified all citations, technical descriptions, analyses, and revisions, and take responsibility for the final content of this work.

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APPENDIX

A OVERVIEW

This supplement documents the implementation and controlled evidence underlying RoadOcc. (1) Sections B–B.2 define the fixed-roadside data and velocity targets used to construct historical addresses. (2) Section C specifies the coupled occupancy, motion, and coverage diagnostics used to assess those addresses and their routed sources. (3) Sections E–E.1 provide the scale-wise objectives and exact DCA–VVE–VDSF execution contract. (4) Sections F–H report efficiency, controlled studies, and scope under the complete model configuration.

B ROADSIDE SENSOR CONFIGURATION

InfraOcc reorganizes synchronized V2X-Real streams from object-level cooperative perception into dense roadside occupancy supervision (Yang et al., 2026). Two fixed infrastructure units each carry one LiDAR and two high-resolution cameras under GPS time synchronization. Calibration maps the four camera streams, two point clouds, and occupancy annotations into a common roadside coordinate system. The four cameras are the only exteroceptive model inputs. LiDAR is used solely for benchmark construction and ground-truth generation. Thus the reported setting remains camera-only despite the multimodal data-collection platform.

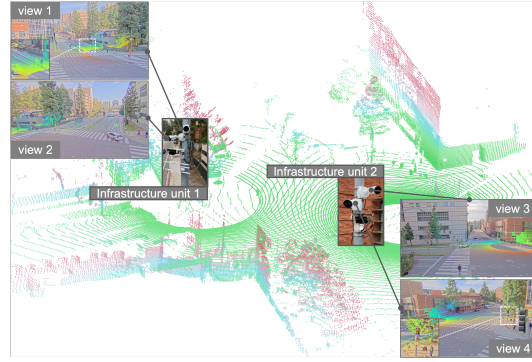


Figure A1: Infrastructure-side sensor configuration. Two fixed units provide four synchronized cameras and two LiDARs calibrated to a common roadside frame.

Roadside benchmarks such as Rope3D and DAIR-V2X also study fixed or cooperative perception (Ye et al., 2022; Yu et al., 2022), but primarily evaluate object-level detection. InfraOcc instead retains a fixed coordinate system for dense occupancy, exposing voxel-level persistence and displacement. This property is essential for testing whether a current voxel retrieves its historical feature from a source-consistent location.

B.1 BENCHMARK SCOPE AND CONSTRUCTION CONTRACT

InfraOcc provides a real-world protocol that combines fixed roadside sensing, temporally continuous dense semantic voxel labels, and camera-only evaluation. Ego-centric occupancy benchmarks assume a moving sensor frame, while existing roadside and cooperative datasets predominantly expose object-level detection or tracking labels. We therefore use the InfraOcc protocol for primary evaluation and complement it with matched P/T/R transfer to two temporal occupancy backbones.

The benchmark contains 290 temporally continuous roadside sequences, with 215 used for training and 75 for evaluation. Each sequence spans approximately 10–24 seconds. Raw streams are captured at 10 Hz and dense occupancy is labeled at 2 Hz, giving the 0.5 s interval used by RoadOcc. Each keyframe contains four calibrated camera views and a semantic occupancy volume over $[-64, 64] \times [-64, 64] \times [-4.8, 1.6]$ m at 0.4 m resolution, or $320 \times 320 \times 16$ voxels. Labels distinguish occupied, free, and unobserved space; unobserved voxels are excluded from supervision and evaluation.

Occupancy construction separates persistent structure from transient objects. A semantic static scaffold is accumulated from registered vehicle-side LiDAR after dynamic-point removal. Dynamic geometry is reconstructed from infrastructure-side LiDAR using annotated object tracklets, box-guided alignment, and local registration. The two sources are recomposed at each keyframe in the fixed roadside frame and voxelized. Ray traversal labels free space before the first observed surface, while voxels behind surfaces or outside valid rays remain unobserved. These LiDAR sources construct supervision only; RoadOcc receives no LiDAR feature as model input.

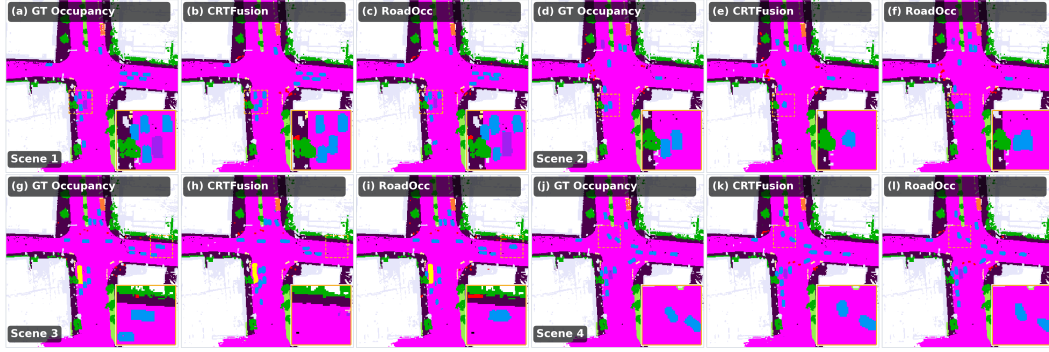


Figure A2: Matched-token occupancy comparison across four scenes. Each triplet shows GT, CRT-Fusion, and RoadOcc in a common BEV frame and semantic palette. The lower-right insets enlarge a representative region that RoadOcc recovers but CRT-Fusion mispredicts.

Qualitative occupancy recovery. Figure A2 compares both methods on identical tokens and coordinates. In the highlighted regions, CRT-Fusion either omits a participant or assigns an incorrect footprint, whereas RoadOcc more closely follows the GT extent and class. The surrounding road structure remains visually unchanged. The gain is therefore concentrated on transient participants rather than obtained by altering the dominant static background.

B.2 VELOCITY-TARGET CONSTRUCTION

InfraOcc occupancy labels do not by themselves define the transport field used by RoadOcc. We therefore generate planar voxel-velocity targets from annotated dynamic boxes and their temporal tracklets. Raw box velocity and adjacent-frame tracklet displacement provide two candidates in the fixed roadside frame.

For track j , let $\mathbf{R}_{\text{rs} \leftarrow \text{raw}}$ transform planar vectors from the annotation frame into the fixed roadside frame, and let $\mathbf{c}_{j,t}^{\text{rs}}$ be its roadside-frame center. The two candidate velocities are

$$\tilde{\mathbf{v}}_{j,t} = \mathbf{R}_{\text{rs} \leftarrow \text{raw}} \mathbf{v}_{j,t}^{\text{raw}}, \quad \mathbf{d}_{j,t} = \frac{\mathbf{c}_{j,t+\Delta t}^{\text{rs}} - \mathbf{c}_{j,t}^{\text{rs}}}{\Delta t}, \quad \Delta t = 0.5 \text{ s}. \quad (11)$$

We denote the selected candidate by $\bar{\mathbf{v}}_{j,t}$. A valid tracklet displacement replaces the raw estimate when the raw estimate is invalid, when their speeds differ by at most 10 m/s, or when their speed ratio is at most two. The corrected velocity is rasterized over dynamic occupancy inside each box, whose vertical extent is expanded by a factor of 1.5 to absorb sparse boundary misalignment. Let $\tilde{B}_{j,t}$ denote this expanded box. The native target assigned to a dynamic voxel is

$$\mathbf{v}_t^{\text{gt}}(\mathbf{x}) = \begin{cases} \mathbf{0}, & \|\bar{\mathbf{v}}_{j,t}\|_2 < 0.8 \text{ m/s}, \\ \text{clip}_{60}(\bar{\mathbf{v}}_{j,t}), & \mathbf{x} \in \tilde{B}_{j,t} \cap \mathcal{O}_t^{\text{dyn}}, \end{cases} \quad (12)$$

where clip_{60} limits the magnitude to 60 m/s. For candidates no faster than 2 m/s, a one-step (0.5 s) future-occupancy check compares zero-motion and transported same-class support. The check is applied only with at least eight support voxels; zero motion is selected when its score exceeds the transported score by at least 0.05. Finally, face-connected component repair propagates valid motion within the corresponding raw dynamic occupancy component, and targets are generated at native, 1/2, 1/4, and 1/8 scales.

Future occupancy is used only to construct offline supervision and evaluation targets. Neither future images nor future features are provided to RoadOcc at training or inference time. The two velocity thresholds serve different roles. The 0.8 m/s threshold denoises the physical target, whereas $\epsilon = 10^{-3}$ m/s is the numerical zero test used by the P/T/R rule after target construction.

C EVALUATION CONTRACT

Aggregate occupancy is dominated by persistent structure and can therefore conceal missed or stale traffic participants. Standard protocols emphasize semantic and geometric IoU (Wang et al., 2023;

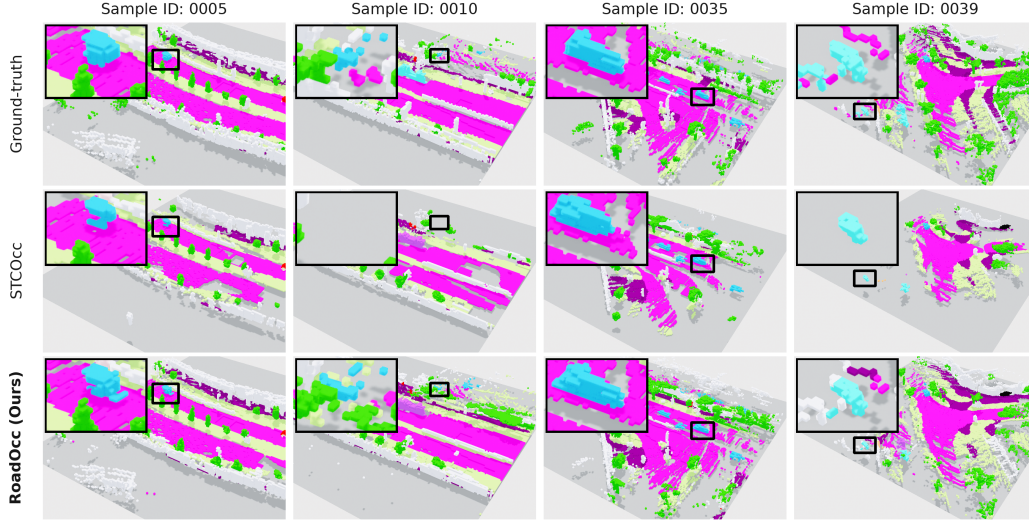


Figure A3: **Qualitative comparison on Occ3D-nuScenes.** We show four validation frames under the official camera-visible mask, semantic palette, voxel grid, and viewing transform. Black boxes and the upper-left insets show the same GT dynamic component in all three rows. Each region is selected where RoadOcc recovers more of that component than the released STCOcc checkpoint.

Tian et al., 2023), whereas dynamic occupancy and flow require motion-aware diagnostics (Ma et al., 2024a; Liu et al., 2025). Table A1 formalizes our evaluation contract on InfraOcc (Xiang et al., 2024; Yang et al., 2026). Dynamic and next-frame IoU measure semantic recovery at the current and flow-advected next-frame locations. Direct MAVE and DSR jointly measure motion accuracy and coverage. Static, geometric, and ray metrics measure whether dynamic recovery preserves the fixed scene. Together, these diagnostics distinguish improved correspondence from reduced dynamic support or retained stale evidence.

Dynamic-object comparison. From left to right, RoadOcc raises recall within the highlighted GT component from 58.5 to 92.5, 0.0 to 87.5, 76.4 to 92.7, and 59.0 to 90.2%, respectively. The insets show the corresponding recovery of missed or truncated traffic participants.

Table A1: Metric and diagnostic scope.

Metric	Support	Failure	Role
Dyn. mIoU $\uparrow$	Dynamic GT voxels	Omission / class error	Dynamic semantic recovery
Static / gIoU $\uparrow$	Static / occupied support	Static-structure damage	Static-scene guardrail
Next Dyn. $\uparrow$	Advected dynamic GT	Displacement error	Next-frame consistency
Direct MAVE $\downarrow$	All dynamic GT voxels	Miss-induced motion bias	Unconditional flow error
TP-MAVE $\downarrow$ / DSR $\uparrow$	Matched semantics / dynamic GT	Conditional-match bias	Matched accuracy / coverage
RayIoU $\uparrow$	Visible evaluation rays	Surface-depth mismatch	Geometric fidelity

Direct MAVE evaluates every valid GT dynamic voxel, whereas TP-MAVE is conditional on a correct semantic match and must therefore be interpreted with DSR. A valid routing improvement should increase dynamic IoU and DSR, decrease Direct MAVE, and preserve the static and geometric guardrails. Reporting these quantities together distinguishes improved semantic correspondence from a lower conditional error caused by reduced semantic support. For completeness, let $\mathcal{D}$ be the set of evaluated dynamic classes, $V_c = \{i : y_i = c\}$ the valid GT voxels of class c , and $V_c^{\text{TP}} = \{i \in V_c : \hat{y}_i = c\}$. Classes without valid support are omitted from the corresponding class macro-average; $\mathcal{D}'$ and $\mathcal{D}''$ denote the supported classes for Direct and TP-MAVE, respectively. The two velocity metrics and semantic recall are

$$\text{Direct MAVE} = \frac{1}{|\mathcal{D}'|} \sum_{c \in \mathcal{D}'} \frac{1}{|V_c|} \sum_{i \in V_c} \|\hat{\mathbf{v}}_i - \mathbf{v}_i\|_2, \quad (13)$$

$$\text{TP-MAVE} = \frac{1}{|\mathcal{D}''|} \sum_{c \in \mathcal{D}''} \frac{1}{|V_c^{\text{TP}}|} \sum_{i \in V_c^{\text{TP}}} \|\hat{\mathbf{v}}_i - \mathbf{v}_i\|_2. \quad (14)$$



Figure A4: Coarse-to-fine occupancy decoding for two representative samples. Columns show predictions from 1/8 scale through full resolution and the corresponding GT. Successive stages sharpen road boundaries and recover traffic participants that remain unresolved at coarse resolution.

Thus MAVE is class-macro while DSR is a micro recall over GT dynamic voxels. For next-frame dynamic IoU, centers of predicted current dynamic voxels are advanced by the predicted planar velocity for 0.5 s, transformed to the next roadside frame, and voxelized. Collisions use deterministic per-class majority assignment before class-macro IoU is computed against next-frame GT. This flow-advected diagnostic is distinct from the same-class radius-one hit score used only in the backwarp visualization.

$$\text{DSR} = \frac{\sum_{c \in \mathcal{D}} |V_c^{\text{TP}}|}{\sum_{c \in \mathcal{D}} |V_c|}. \quad (15)$$

D OCC3D-NUScENES TRANSFER

Table A2 compares representative camera-only temporal methods under the mask-trained Occ3D-nuScenes protocol (Tian et al., 2023; Li et al., 2023c; Chen et al., 2025a; Li et al., 2024; Ma et al., 2024b; Liao et al., 2025). Dynamic mIoU averages the eight traffic-participant classes, while static mIoU averages the remaining nine occupied classes. We compute both

Table A2: Mask-trained comparison on Occ3D-nuScenes. All metrics are percentages.

Method	BB	Res.	mIoU	Dynamic mIoU	Static mIoU
FB-Occ	R50	256×704	39.10	33.75	43.89
ViewFormer	R50	256×704	41.90	35.00	47.93
COTR	R50	256×704	44.50	38.60	49.63
ALOcc	R50	256×704	41.40	35.40	46.73
STCOcc	R50	256×704	44.60	39.01	49.58
RoadOcc	R50	256×704	45.01	39.80	49.64

group metrics from published per-class IoUs when available. ALOcc directly reports dynamic mIoU; its static mIoU follows from its reported dynamic and overall mIoUs. Before rounding, the three aggregates satisfy $\text{mIoU} = (8 \text{ Dyn.} + 9 \text{ Sta.})/17$. RoadOcc leads all three aggregates, improving STCOcc by 0.41 overall, 0.79 dynamic, and 0.06 static mIoU points. The larger dynamic gain shows that motion-aware routing transfers without sacrificing static-scene accuracy.

E TRAINING OBJECTIVES

RoadOcc assigns each loss to one decision in the DCA-VVE-VDSF chain. Camera-depth supervision uses weight 0.5. Semantic occupancy is supervised from full resolution to 1/8 scale with weights $[1.0, 0.5, 0.25, 0.125]$, while the static, dynamic, and DCA-candidate auxiliaries each use 0.5. VVE receives class- and speed-balanced robust flow supervision at all four scales with weight 0.1. Its local-connectivity term also uses 0.1 and starts after epoch 3 to limit noisy early correspondences. The P/T/R route loss spans the native, 1/2, 1/4, and 1/8 scales with relative weights $[1, 0.5, 0.25, 0.125]$, normalized under $\lambda_{\text{route}} = 0.5$. Each scale constructs the semantic-support target using its physical cell size. Equal-class-weight cross-entropy is augmented from epoch 2 by a

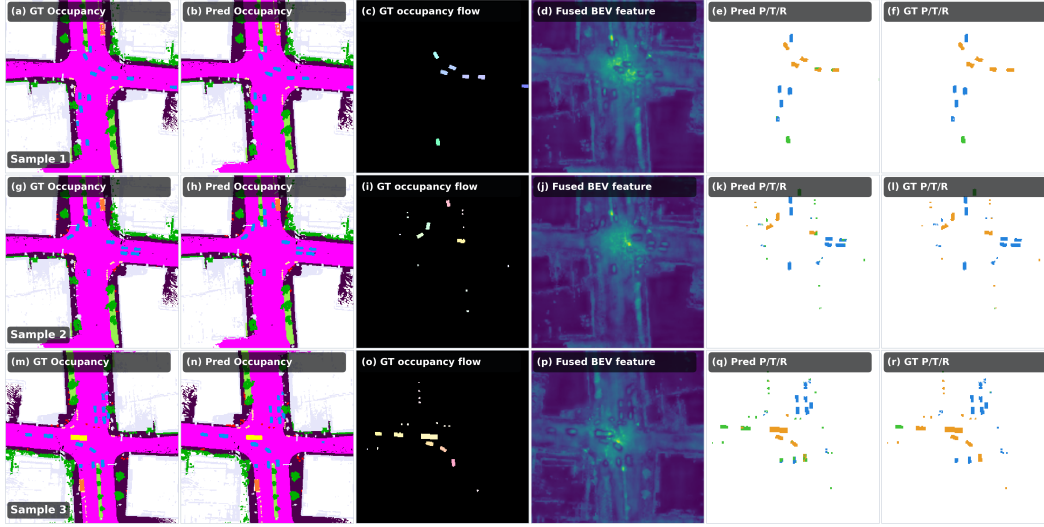


Figure A5: Native-scale P/T/R routing across three samples. Predicted and GT routes share GT dynamic support. Blue, orange, and green denote Persist, Transport, and Refresh, respectively.

soft-overlap term of weight 0.5 with state weights $[1, 1, 0.25]$. Each executed routing field therefore receives resolution-matched supervision.

E.1 CANDIDATE, MOTION, AND ROUTING CONTRACT

DCA score. Let π_{cur} and π_{sta} denote current and static-reference class probabilities, and let $\mathcal{C}_{\text{sta}}^+$ contain the static classes together with empty class. The executed change and dynamic scores are

$$\delta(\mathbf{x}) = \frac{1}{|\mathcal{C}_{\text{sta}}^+|} \sum_{c \in \mathcal{C}_{\text{sta}}^+} |\pi_{\text{cur},c}(\mathbf{x}) - \pi_{\text{sta},c}(\mathbf{x})|, \quad (16)$$

$$p_{\text{dyn}}(\mathbf{x}) = \sum_{c \in \mathcal{C}_{\text{dyn}}} \pi_{\text{cur},c}(\mathbf{x}), \quad (17)$$

$$c(\mathbf{x}) = \frac{\max\{\delta(\mathbf{x}), p_{\text{dyn}}(\mathbf{x})\}}{\max_{\mathbf{y}} \max\{\delta(\mathbf{y}), p_{\text{dyn}}(\mathbf{y})\} + 10^{-6}}. \quad (18)$$

DCA ranks this normalized score. Before fixed top- K selection, VDSF uses $\text{clip}(c + 0.5e, 0, 1)$, where e is nonempty support and hence $\eta = 0.5$ in Eq. 6. For an anchor $\mathbf{r} = (\mathbf{x}, z_{\mathbf{r}})$, $c(\mathbf{r})$ is the score of the corresponding voxel-height bin and is shared across cameras before visibility masking. Let $d_{n,\mathbf{r}}$ be its projected depth, $\hat{d}_n$ the winning predicted depth bin, and $\Delta_d = 1$ m the bin width. The executed depth-consistency factor is

$$\beta_{n,\mathbf{r}} = \exp \left[-\frac{\sigma^2}{2} \min_{\xi \in \{-\Delta_d, +\Delta_d\}} |d_{n,\mathbf{r}} - (\hat{d}_n + \xi)|^2 \right], \quad \sigma = 2. \quad (19)$$

Deformable-attention weights are first softmax-normalized over sampled points, then multiplied by β . They are not renormalized afterward; camera outputs are averaged over cameras retaining at least one visible anchor.

Recursive occupancy decoding. Figure A4 shows the progressive output of the multi-scale decoder. Coarse predictions establish scene layout, while finer stages restore object boundaries and sparse traffic participants before the full-resolution output.

VVE correspondence range. At scales $1/8$, $1/4$, and $1/2$, the 5×5 correlation volume ($r = 2$) spans ± 6.4 , ± 3.2 , and ± 1.6 m on the native 0.4 m grid. Over the 0.5 s interval, the equivalent speed ranges are twice these values.

Each correlation operand is normalized to unit ℓ_2 norm with $\epsilon = 10^{-6}$. The resulting volume has 25 channels and is concatenated with current and historical BEV embeddings, processed by two

3×3 convolution blocks, and decoded as a two-dimensional velocity residual in m/s. The window bounds only the explicit matching evidence. The coarse velocity is added to each finer prediction and the final continuous residual is not clipped, so it does not impose a hard bound on predicted speed. The GT-flow study in the main paper isolates the value of transport addresses across speed ranges. In the temporal blend, the history branch receives weight $0.5 \mathbf{1}_{\text{hist}} \max_z p_{\text{dyn}}(\mathbf{x}, z)$; this is current dynamic support gated by history validity, rather than a calibrated measure of historical correspondence.

Native P/T/R target. For a current dynamic voxel, the Transport source is

$$\mathbf{x}_{\text{src}} = \text{round}\left(\mathbf{x} - \frac{\Delta t \mathbf{v}_t(\mathbf{x})}{0.4 \text{ m}}\right), \quad \Delta t = 0.5 \text{ s.} \quad (20)$$

Persist tests the fixed-coordinate address and Transport tests $\mathbf{x}_{\text{src}}$. Support means that the nearest previous GT contains the same semantic class at the same height within a 3×3 XY neighborhood (L_∞ radius one). Support is Boolean. One or more matching voxels suffice, with no instance-level tie-breaking. Out-of-grid or unsupported candidates are assigned Refresh rather than removed. The target follows InfraOcc’s voxel-level semantic supervision and therefore tests class-consistent support without requiring instance identities.

Coarse-to-fine FIFO routing. At each decoder scale, a shared controller projects current features and the nearest fixed-coordinate history to eight channels, adds embeddings of detached VVE flow (normalized by 10 m/s) and detached upsampled coarse P/T/R, and predicts three route logits after a $3 \times 3 \times 3$ refinement. A lightweight native head combines upsampled 1/2 context, native flow, and the finest P/T/R distribution for full-resolution routing. Each scale shares its route distribution across valid FIFO slots, whose ages and cached VVE fields determine source-consistent inverse-trajectory Transport addresses. Routed slots are concatenated before fusion. Invalid slots are masked and queues reset at scene boundaries. Fusion uses soft routes, with argmax only for diagnostics.

F CAPACITY AND EFFICIENCY CONTEXT

Table A3 reports parameters, synchronized model latency measured on a single NVIDIA RTX A6000 with batch size 1, and occupancy accuracy for the reproduced temporal configurations. Although RoadOcc contains more parameters than STCOcc, it reduces latency from 333.44 to 253.25 ms because DCA, VVE, and P/T/R operate on sparse or low-resolution features, while the main temporal path uses shorter histories and smaller token budgets. This efficient routing also improves all-class and dynamic mIoU by +4.44 and +4.71 points, respectively.

Class-wise improvement. Figure A6 resolves the official result in Table 2 into dynamic and static classes. Relative to STCOcc, RoadOcc improves five of the six dynamic classes, led by Bike at +10.47 and Bus at +6.68, while Pedestrian changes by -0.19 . Every static class also improves. The category pattern therefore connects the aggregate gain to both parts of the routing objective. Corrected addresses recover transient participants, while source selection preserves the persistent roadside scene.

G CONTROLLED DESIGN STUDIES

All studies hold the backbone, data protocol, optimizer, training schedule, scene-boundary queue reset, and VDSF configuration fixed. Unless varied explicitly, the decoder stages use $K = [2000, 512, 128]$ sparse tokens and retain

Table A3: Accuracy–efficiency comparison.

Setting	Params.↓	Lat.↓	All↑	Dyn.↑
STCOcc '25	60.47	333.44	60.85	27.66
RoadOcc (ours)	78.69	253.25	65.29	32.37

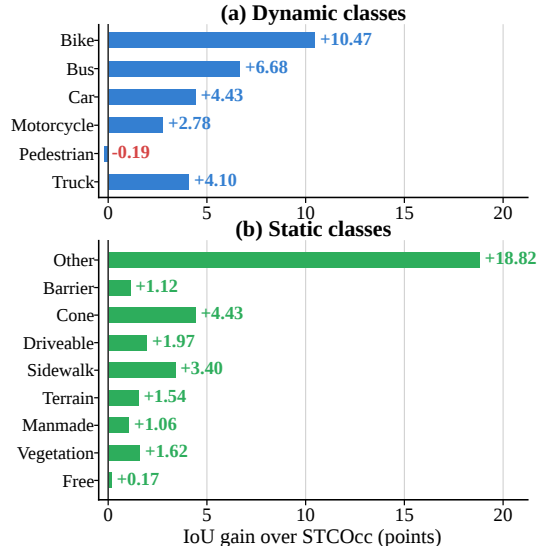


Figure A6: Class-wise IoU gains over STCOcc.

[8, 4, 2] history samples, respectively. The main paper reports the matched address and P/T/R controls. The supplementary studies below provide non-overlapping evidence on VVE training, route prediction, sparse allocation, and address construction.

Temporal-gap zero-shot generalization. Table A4 evaluates checkpoints trained only at the 0.5 s cadence without fine-tuning. Phase-offset streams retain all 1,970 validation inputs, while the same 1,520 anchors with valid past and future 1.5 s context are scored at every gap. The elapsed time is recovered from timestamps and used by temporal transport.

Table A4: Temporal-gap zero-shot evaluation on matched anchors. Both checkpoints are trained only at a 0.5 s interval.

Method	Dyn. $\uparrow$			Nxt. $\uparrow$		
	0.5 s	1.0 s	1.5 s	0.5 s	1.0 s	1.5 s
STCOcc (Liao et al., 2025)	28.04	28.26	28.58	25.21	20.90	19.12
RoadOcc (ours)	33.71	33.57	33.49	30.23	25.18	22.78

RoadOcc preserves a 4.91-point Dyn. advantage and a 3.65-point Nxt. advantage even when the test gap triples. VVE represents displacement through a physical velocity field, while P/T/R selects fixed-coordinate, transported, or current evidence. Their combination therefore remains effective when elapsed time changes rather than depending only on the observed 0.5 s transition pattern.

VVE training design. The main-paper control establishes the value of reading from a corrected address. Figure A7 follows a progressive sequence that introduces multi-scale flow, candidate-map guidance, connectivity regularization, and flow-addressed history retrieval. Dyn. improves monotonically while Direct MAVE decreases across this progression. Overall, Dyn. rises by +3.63 and Direct MAVE falls by 0.271. The final flow-addressed read contributes another +1.65 Dyn. while Direct MAVE decreases by only 0.021, indicating that its main benefit comes from retrieving useful history.

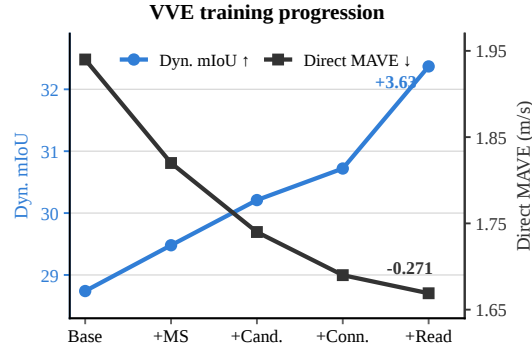


Figure A7: Dynamic occupancy and motion-error progression under the VVE training signals.

Supported-route analysis. Table A5 audits the two historical routes on valid GT dynamic voxels. Persist emphasizes precise reuse of fixed-coordinate evidence, while Transport provides broader coverage after displacement. Since VDSF fuses soft probabilities over all source alternatives, these argmax statistics characterize history reuse rather than the complete fused output. The end-to-end effect of routing is evaluated separately through Dyn., Nxt., Direct MAVE, and DSR.

Qualitative routing behavior. Figure A5 relates each route to observable scene evidence. Persist coincides with aligned objects, Transport follows nonzero displacement, and current evidence is selected when neither historical source is supported. Because predicted and GT routes are rendered on identical dynamic support, their differences directly reflect source choices. The visualization therefore audits semantic source routing independently of occupancy detection.

Sparse-token budget. VDSF processes only the top- K selected voxels. The token study

tests whether the reported behavior depends on a larger compute budget or on identifying the right locations. All other model and routing settings remain fixed while K is scaled at every decoder stage. Increasing the small allocation to the default improves Dyn. by 0.63 and DSR by 0.83. Doubling the default allocation changes Dyn. by only 0.02, dMAVE by 0.004, and DSR by 0.07, while

Table A5: Supported historical-route statistics on valid GT dynamic voxels. Percentages are computed against the full routing population.

State	Target	Pred.	Precision	Recall
Persist	52.34	38.84	95.50	70.87
Transport	44.30	50.40	72.98	83.03

Table A6: Sensitivity to sparse-token allocation.

Budget K	Dyn. $\uparrow$	Nxt. $\uparrow$	dM $\downarrow$	DSR $\uparrow$
[1000, 256, 64]	31.74	30.96	1.716	60.28
[2000, 512, 128]	32.37	31.72	1.669	61.11
[4000, 1024, 256]	32.39	31.69	1.665	61.18

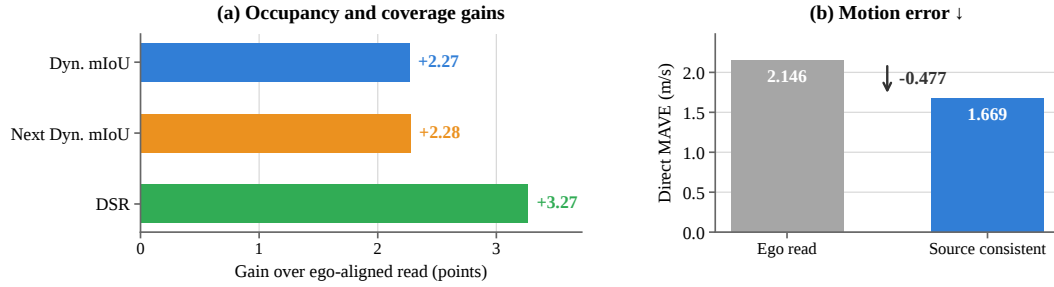


Figure A8: Measured advantage of source-consistent address construction over ego-aligned reading. Values are derived from the matched controlled settings in Table A7.

next-frame IoU fluctuates by -0.03 . This saturation shows that the reported result does not rely on continually expanding the sparse-token budget.

Velocity-address construction. Velocity is useful for fusion only when it points to evidence that explains the current target. Table A7 therefore compares a fixed-coordinate read, a target-vector backtrace, and the complete source-consistent rule under an otherwise fixed P/T/R router. This isolates address construction from subsequent source acceptance and complements flow-based warping in prior occupancy models (Liu et al., 2025; Liao et al., 2025).

Target-vector backtracing improves Dyn. by 1.31 and reduces dMAVE by 0.442 over the ego-aligned read. Enforcing source consistency contributes a further 0.96 Dyn., 1.06 Nxt., and 1.29 DSR while reducing dMAVE by 0.035. The paired occupancy and coverage gains show that the refined address is useful for retrieval rather than only for velocity regression.

Table A7: VVE address control.

Address	Dyn. ↑	Nxt. ↑	dM ↓	DSR ↑
Ego-aligned read	30.10	29.44	2.146	57.84
Target-vector trace	31.41	30.66	1.704	59.82
Source-consistent	32.37	31.72	1.669	61.11

Stepwise address benefit. Figure A8 makes the controlled progression in Table A7 explicit. Target-vector tracing corrects motion-aware retrieval, while source consistency adds a further gain by ensuring that the transported source remains semantically admissible for the current voxel. Across the full progression, Dyn., Next Dyn., and DSR rise by 2.27, 2.28, and 3.27 points, respectively, while Direct MAVE falls by 0.477.

H IMPLEMENTATION AND SCOPE

The production configuration uses four cameras, a ResNet-50 backbone, three decoder stages, a 0.4 m voxel grid, and a 0.5 s sampling interval. Models are optimized with AdamW for 24 epochs. Controlled variants share the same initialization policy, optimizer, schedule, scene-boundary queue reset, and InfraOcc evaluator. Only the designated factor is changed.

P/T/R operates on semantic voxel support, matching the InfraOcc label space. Nearby same-class objects can share support within the local radius; when history has no supported source, Refresh uses the current-frame feature. The route target therefore encodes semantic evidence admissibility rather than instance identity. We report dynamic IoU, Direct MAVE, and DSR jointly so that semantic recovery, motion error, and support coverage remain visible beyond the GT-conditioned route diagnostic.